Sentinel

Adaptive time-series anomaly detection

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Challenge 2

Two statistical checks flag unusual events as readings arrive. Each alert includes a short explanation.

Event-F1 0.691

2.8 µs per point

Problem and motivation

- A time series is a sequence of readings, such as CPU usage, traffic, or sales.
- Sentinel must flag unusual behaviour **as each reading arrives**, using only earlier readings.
- Missed incidents matter. Too many false alerts waste people's attention.
- Each result has a score, a NORMAL or ANOMALOUS label, and an explanation.

Goal: accurate event alerts with low delay and low runtime.

Data and preparation

NAB category	Files	Anomaly windows
Artificial	11	6
AWS CloudWatch	17	30
Ad Exchange	6	14
Known Cause	7	19
Traffic	7	14
Tweets	10	33
Total	**58**	**116**

365,558
readings

Learning period

First 15% of each file, up to 750 readings. No alerts or scoring during this period.

Each file stays in its original order; repeated or out-of-order timestamps are handled.

System design

1	**Reading arrives**	Time and value
2	**Check input**	Clean invalid readings
3	**Two detectors**	RobustZ + LaggedResidual
4	**Decide**	Adaptive thresholds + new high/low
5	**Manage event**	Confirm, join, space alerts
6	**Return result**	Score, label, explanation

Every decision uses the current and earlier readings only.

Core algorithm

Part	What it does
RobustZ	Checks whether a reading is far from its usual level, allowing for normal variation.
LaggedResidual	Compares it with the average of 8 readings from roughly 256 points earlier.
Adaptive thresholds	Each detector sets its own alarm level using high learning-period scores.
New high / low	Flags a value beyond the range seen so far by the chosen margin.

RobustZ starts with the median of learning-period readings. It also measures the usual spread, so being above the median alone does not trigger an alert.

Examples from the current default

RobustZ: CVS tweet count reached 10 versus an expected 0.41. It was far above its usual level.

Spatial-extreme flag: Ambient temperature read 82.29 versus the earlier high of 80.52, crossing the new-high margin.

Both are inside labelled anomaly windows. The spatial flag is a separate new-range check alongside the two statistical detectors.

Experiment history and tuning

We measured each change before choosing one global default.

1. Compared simple baselines and detector signals on NAB.
2. Searched detector combinations, lag lengths, and window lengths.
3. Tested the new-high / new-low margin and event grouping rules.
4. Selected RobustZ + LaggedResidual, an 8-reading older average, a 256-reading lag, and a 5.75% margin.

Validation: rotate which NAB category is held out when selecting settings.

Evaluation method

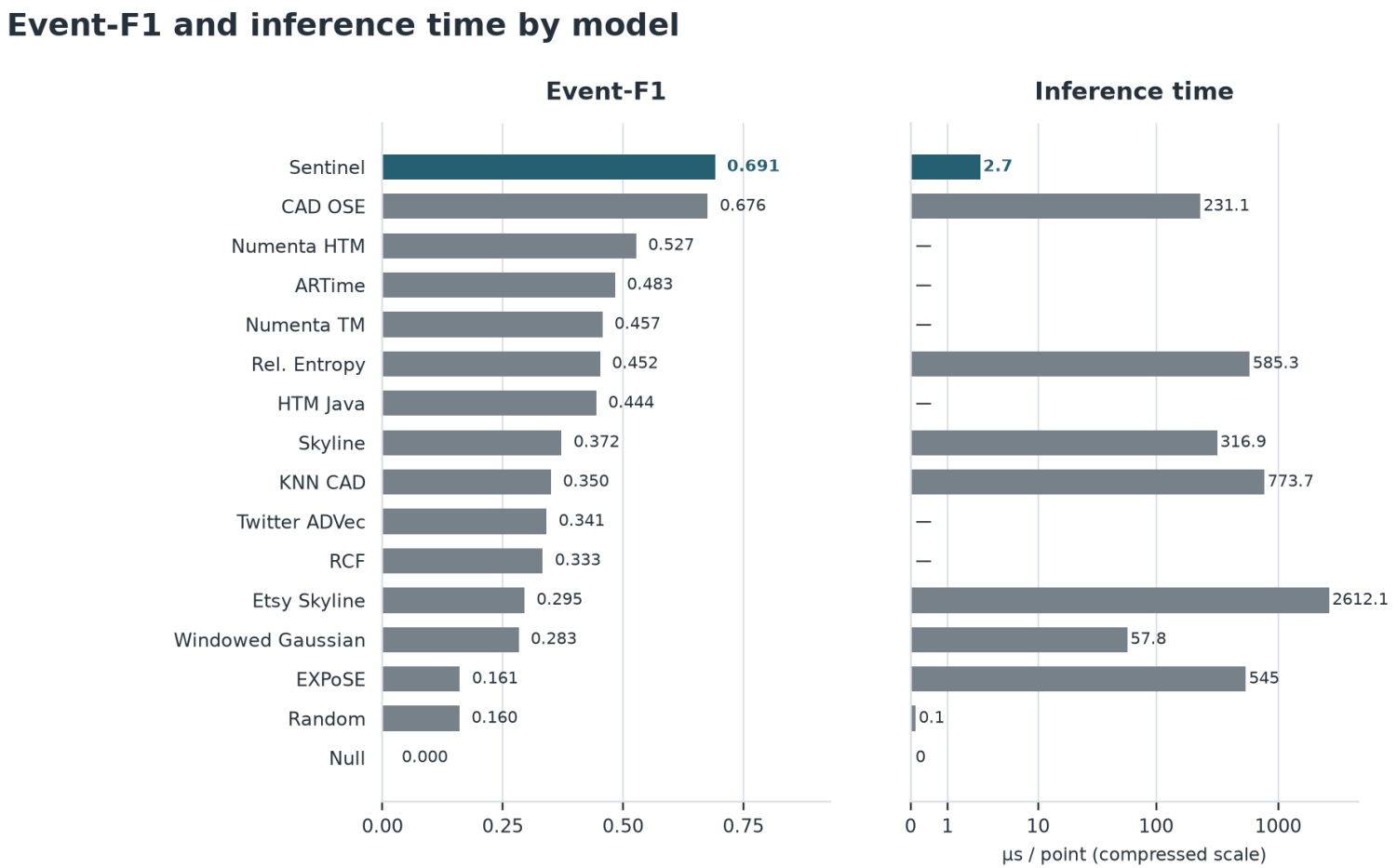
An alert inside a labelled anomaly window counts as a true positive. An alert outside all windows is a false positive. A window without an alert is a false negative.

Current default	Value	Across 58 files	Count
Precision	0.731	True positives	76
Recall	0.655	False positives	28
Event-F1	**0.691**	**False negatives**	**40**
Official NAB score	58.80	Median delay	100 readings

Delay is measured from window start to the first alert, among detected windows only.

Model	Windows found / 116	Median position
Sentinel	**76**	**49.7%**
CAD OSE	92	49.7%
Numenta HTM	96	49.6%
ARTime	106	49.9%
Windowed Gaussian	73	48.8%

Event-F1 and inference time



Sentinel default

0.691 event-F1

2.7 µs per point

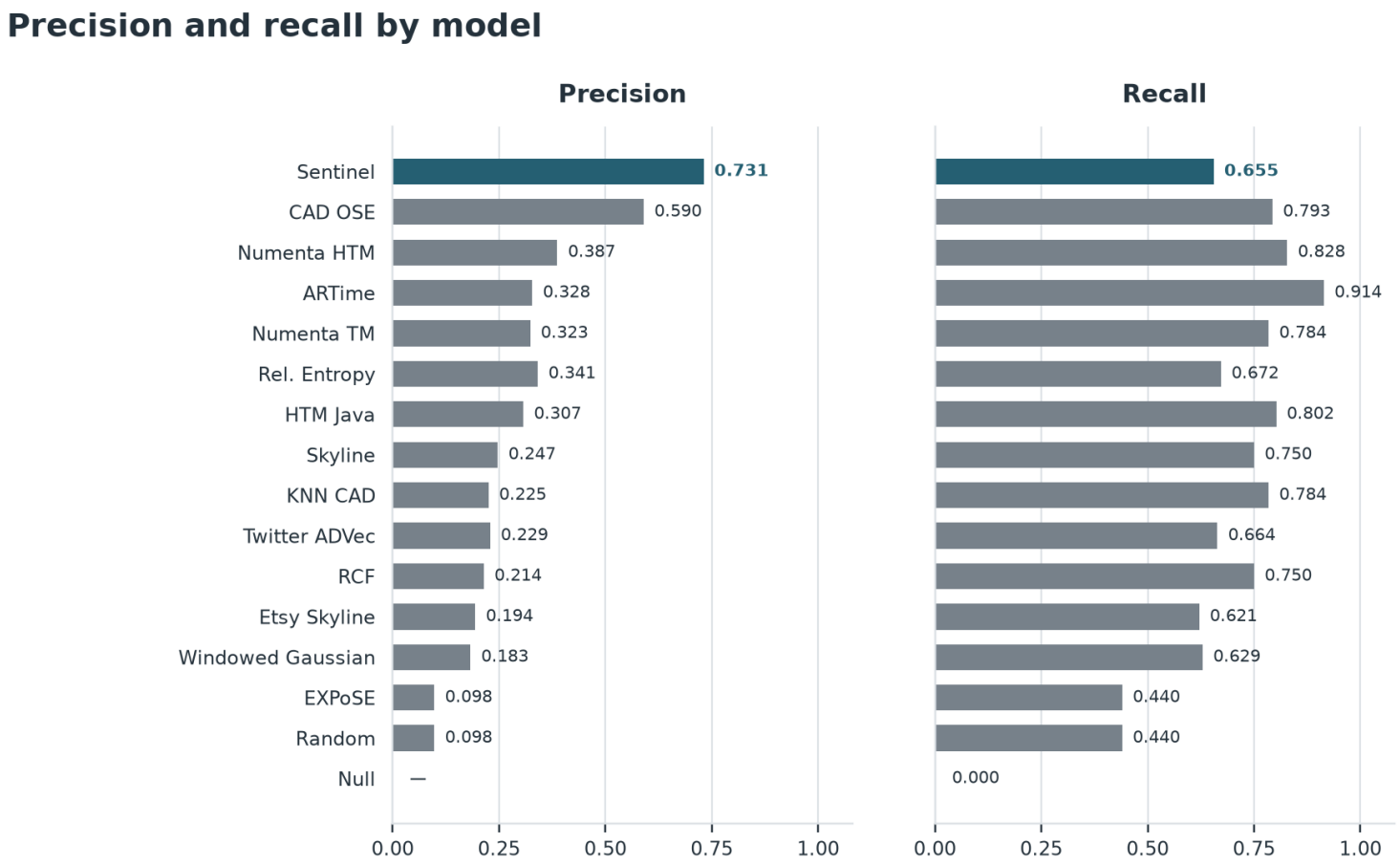
CAD OSE

0.676 event-F1

231 µs per point

Published models use NAB's stored scores and thresholds. Same-stream runtimes; time axis compressed, with actual µs in labels. A dash means no local runtime.

Precision and recall



Sentinel default

Precision 0.731

Recall 0.655

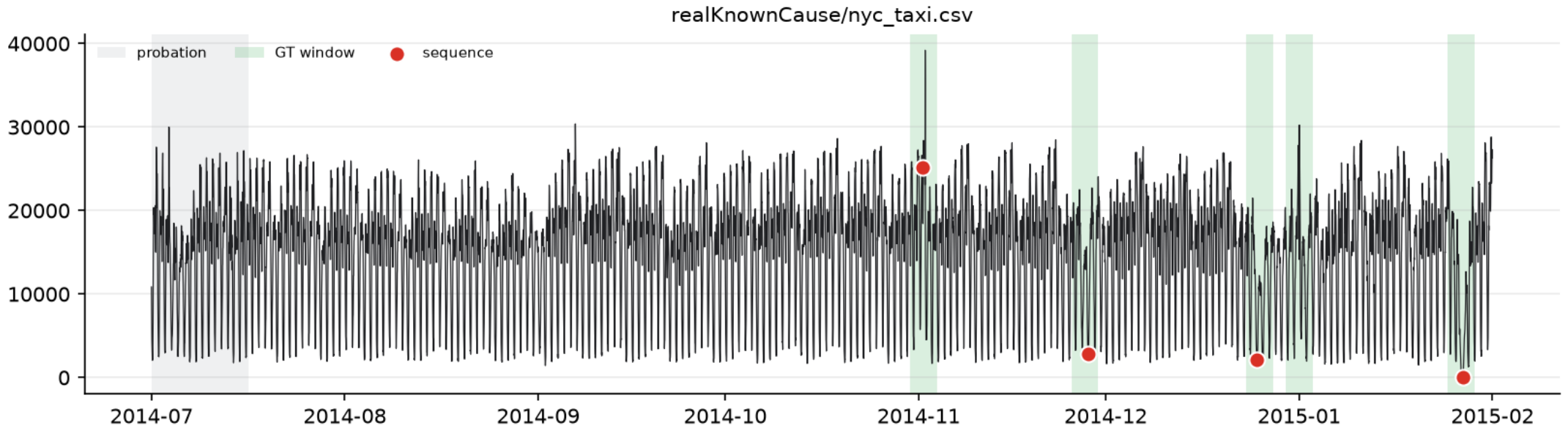
CAD OSE

Precision 0.590

Recall 0.793

Published models use NAB's stored scores and thresholds. More windows found can come with more false alerts. A dash means undefined precision.

Four NYC taxi alerts



Alert	Reading	Older average	Why LaggedResidual flagged it
1 Nov 2014, 13:30	25,115	2,602	Much higher than the earlier level
28 Nov 2014, 06:00	2,870	24,890	Much lower than the earlier level
25 Dec 2014, 06:00	2,144	26,316	Much lower than the earlier level
27 Jan 2015, 06:00	69	22,673	Nearly zero compared with the earlier level

Each older average uses 8 readings roughly 256 points earlier. Four of five labelled windows found; no false events.

Limits and next steps

- **40 of 116** labelled windows are missed. Changes in daily patterns can be hard to spot when readings remain within the usual range.
- The default was selected and measured on public NAB data. Performance on hidden data is unknown.
- Next: detect pattern changes while keeping false alerts and runtime low.

Conclusion

Two causal statistical checks produce clear alerts at 0.691 event-F1 and about 2.8 µs per point.